



Sampling Filter dedicated for Dioxins

Instruction manual

Instruction manual can be downloaded from Miura HP
http://www.miuraz.co.jp/e_science/doc/index.html

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X001-000-5114-0

ダイオアナフィルタをお買い上げいただきありがとうございます。この製品を安全に正しくご使用いただくため、お使いになる前にこの取り扱い説明書をお読みになり、十分にご理解ください。

使用上の注意

- ダイオアナフィルタは、排ガス試料中のダイオキシン類を採取するためのものです。これ以外の用途に本製品を使用しないでください。
- ダイオキシン類の採取には、ダイオアナフィルタ専用の器具をご使用ください。
- ダイオアナフィルタの取扱いは、ダイオキシン類に汚染されていない手袋、マスク等を着用した上で行ってください。
- 強い力を加えるとダイオアナフィルタが破損する恐れがあります。取り扱いにはご注意ください。
- 開栓後はダイオアナフィルタへのほこりや汚れ等の付着を避け、すみやかにご使用ください。
- 開栓後はダイオアナフィルタの汚染および損傷については一切の責任を負いません。
- 高温、多湿を避けて保管してください。
- 廃棄する場合は「廃棄物の処理および清掃に関する法律」に従って廃棄してください。
- その他、ご不明な点については下記までお問い合わせください。

お問い合わせ先

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Dioana Filter use conditions

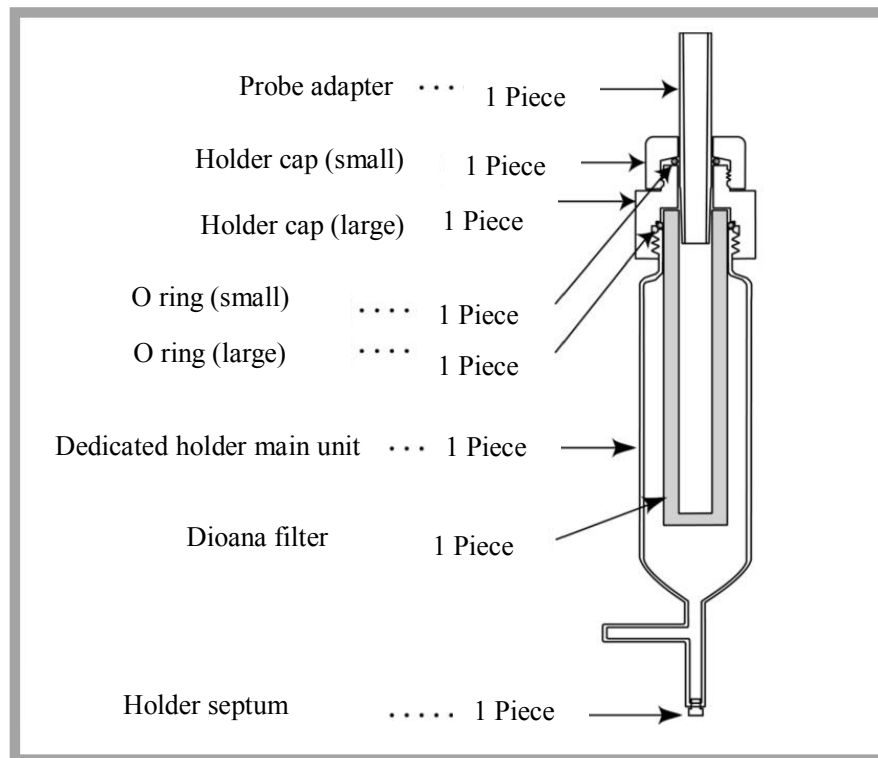
Item	Use conditions	Instructions
Collecting part temperature	The range is between moisture incondensable temperature and 120°C	There is possibility dioxins secondary formation and dissolution if temperature is over 120°C Preheat filter and absorbing collecting parts before starting sampling to minimize condensation at absorbing collecting parts when sampling starts. Monitor temperature after Dioana filter during sampling, and confirm that it is not less than 120°C and moisture is not condensed just before Dioana filter.
Dust	Collect dust to attach such as thimble filter before	
Sampling spike		Add sampling spike to 2 cm under from opening of Dioana filter and use after vaporizing solvent immediately. Use acetone or hexane as additional liquid and add not less than 200 μ L.

The range of application of Dioana Filter conditions

Item	The range of application	Instructions
Objective gas	General / Industrial incinerator flue gas	
Objective substance	Dioxins (PCDDs, PCDFs, DL-PCBs)	
Moisture content	Not less than 36%	
Actual average CO concentration	Not less than 670 ppm	
Passing flow rate	33L-wet/min at not less than 120°C	Passing flow rate when Dioana filter is kept warm at 120°C For example, passing flow rate is about 18L-dry/min at gas meter when moisture content is 30% and dry gas meter temperature is 30°C.
Sampling volume	Not less than 3.8m ³ N-dry	

- It is not confirmed result which it is impossible to use in range of except above condition.
- Refer to Miura technical report (http://www.miura.co.jp/e_science/doc/index.html) about revalidation confirming test result.

Dioana Filter parts name



Dioana Filter specification

Quality of material	Fibriform alumina
Size	$\phi 18.8 \times 110L$
Pressure loss [kPa(mmH ₂ O)] ¹⁾	< 2 (200)
Dioxin content (ng-TEQ) ²⁾	< 0.002

1) Pressure loss: value when aeration speed is 30L/min (120°C air)

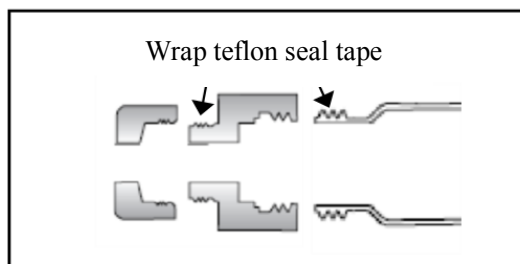
2) By Method for determination of tetra-through octachlorodibenzo-p-dioxins, tetra-through octachlorodibenzofurans and dioxin-like polychlorinatedbiphenyls in stationary source emissions JIS K 0311:2008

Method of mounting Dioana Filter

1. Preservation of screw parts of holder cap (aluminum)

Wrap teflon seal tape to the screw parts of holder cap (large) and single purpose holder body to preserve them.

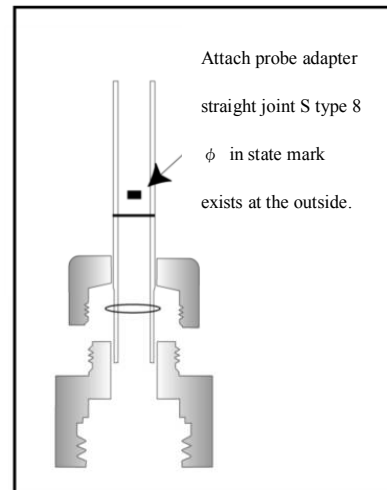
- Holder cap is made of aluminum. Preserve jammed by contaminant and eliminate an unnatural force.



2. Holder cap assembly

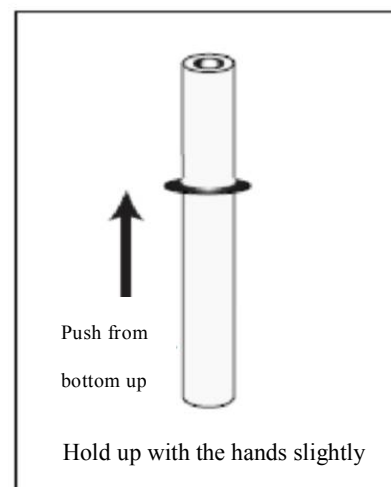
Attach O ring (small) to holder cap (small) through probe adapter. Fasten large cap and small in state which line of probe adapter exists at the outside of top of holder cap (small).

If you want to arrange location of probe adaptor again, do it after loosening cap (large and small).



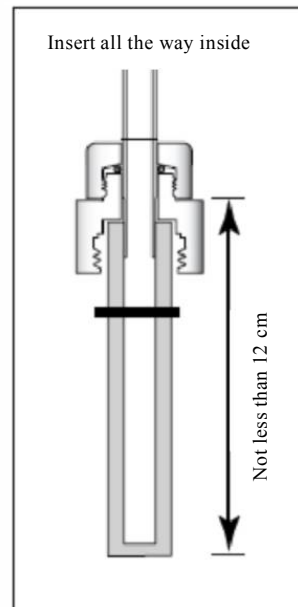
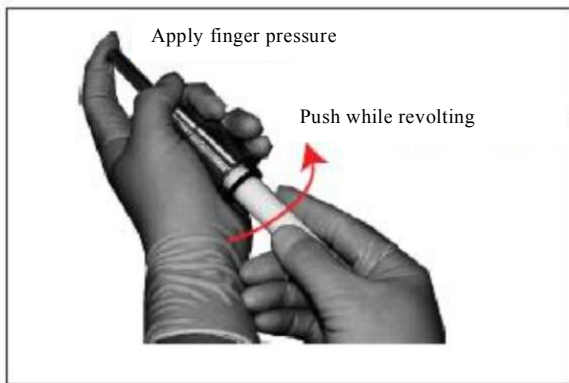
3. Dioana Filter assembly

Attach O ring (large) to Dioana Filter in advance like right figure. Attach O ring (large) from the bottom (closing orifice) invariably. Pull from bottom when O ring is put off.



4. Dioana Filter attachment

Insert Dioana Filter to holder cap which built-up in section 2 up until bottom while revolting attached O ring Dioana Filter.

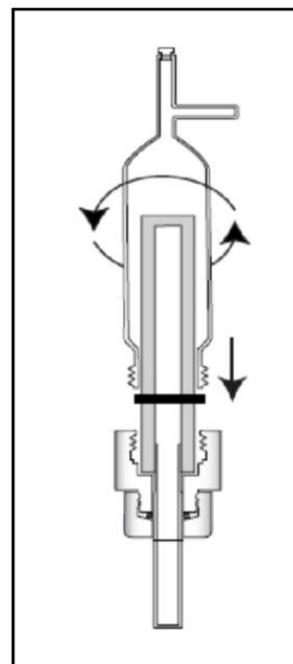


Confirm length between upper of holder cap (large) and top of Dioana Filter is not less than 12 cm.

- Make sure the handling, because it is glassware.

5. Dedicated holder assembly

Stand holder cap downside and upright, and clench while revolting dedicated holder slowly along pitch. Clench strongly until feeling dedicated holder attached to bottom of holder cap.



6. Holder septum attachment

Attach septum to sectional nozzle at extension of Dioana Filter.

- Holder septum is to insert temperature sensor.

7. Confirmation and instructions

Confirm Dioana Filter attachment situation

Confirm following after assembling dedicated holder. And confirm again before use when dedicated holder is transported in assembled situation.

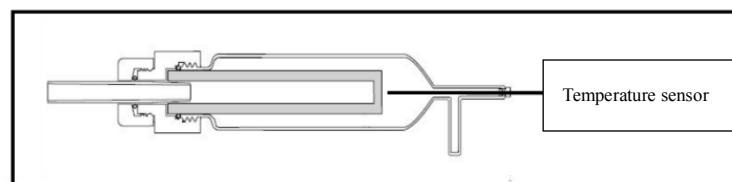
- Holder caps (large and small) are strong clenched.
- The length between upper of holder cap (large) and top of Dioana Filter is within 12 cm.
- Dioana filter doesn't contact with dedicated holder.

Confirm before sampling

Check leak before sampling invariably. Confirm such as horses, holder septum and holder cap when there is leakage.

Instructions about location of temperature measurement

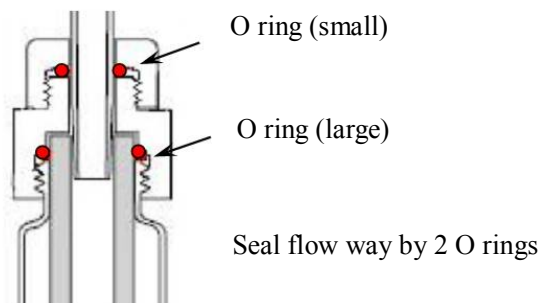
Measure flue gas temperature to insert temperature up to near Dioana Filter. Don't measure temperature because temperature is easy to fall in flue gas temperature in branching tubule.

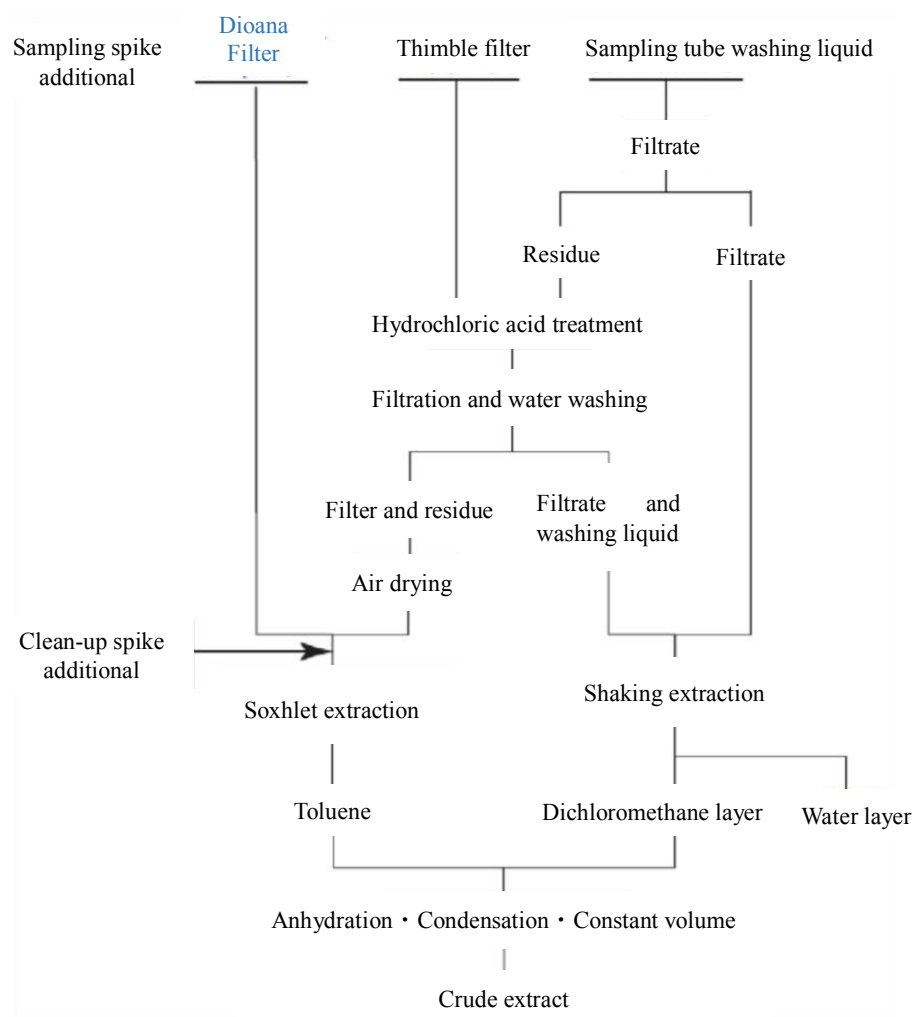


O ring periodic replacement

Troubles such as O ring flaw and hardening cause fixation and leakage. Prevent troubles to replace proudly if it is no O ring trouble.

Holder O ring set(X001-000-0241-4)





High speed solvent extraction condition (example)

Extraction solvent	Toluene
Temperature	150 °C
Pressure	13.8 MPa (2000 psi)
Heating time	7 minutes
Static time	2 minutes
Flash capacity	70 %
Purge time	60 seconds
Static cycle time	5 times
Repeat	2 times